01_data_exploration

March 27, 2026

1 Analysis of the data base

The data base is composed of all matches of 5 seasons : from 2020/2021 to 2024/2025 will be used to test our model

Source : understat.com

[32]:
```
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
import seaborn as sns
```

[33]:
```
df = pd.read_excel('/Users/baptistesmac/Documents/Formation finance/ML ligue 1/
↪data/raw/understat2_ligue1_matches.xlsx')
df.head()
```

[33]:
```
    season       date  home_team            away_team  home_goals  away_goals  \
0  2016/17 2016-08-12  SC Bastia  Paris Saint Germain           0           1
1  2016/17 2016-08-12     Monaco             Guingamp           2           2
2  2016/17 2016-08-13   Bordeaux        Saint-Etienne           3           2
3  2016/17 2016-08-13       Caen              Lorient           3           2
4  2016/17 2016-08-13       Metz                Lille           3           2

   home_xg  away_xg  home_shots  away_shots  home_shots_on_target  \
0   0.8284   0.6649          10           9                     0
1   2.4964   0.6990          18           9                     4
2   1.1659   1.5013          11          14                     6
3   2.2728   1.9864          16           8                     7
4   2.3171   1.8778           7          17                     3

   away_shots_on_target  home_ppda  away_ppda  home_deep  away_deep Result
0                     6    12.6667     4.1154          0          3      A
1                     5     2.6053    12.2857          9          2      D
2                     5    11.5000     5.0313          3          7      H
3                     2     8.9583    12.6667          8          7      H
4                     4    11.6667    15.2143          4          4      H
```

[34]:
```python
df.info()
print(df.isnull().sum())
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3412 entries, 0 to 3411
Data columns (total 17 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   season                3412 non-null   object
 1   date                  3412 non-null   datetime64[ns]
 2   home_team             3412 non-null   object
 3   away_team             3412 non-null   object
 4   home_goals            3412 non-null   int64
 5   away_goals            3412 non-null   int64
 6   home_xg               3412 non-null   float64
 7   away_xg               3412 non-null   float64
 8   home_shots            3412 non-null   int64
 9   away_shots            3412 non-null   int64
 10  home_shots_on_target  3412 non-null   int64
 11  away_shots_on_target  3412 non-null   int64
 12  home_ppda             3412 non-null   float64
 13  away_ppda             3412 non-null   float64
 14  home_deep             3412 non-null   int64
 15  away_deep             3412 non-null   int64
 16  Result                3412 non-null   object
dtypes: datetime64[ns](1), float64(4), int64(8), object(4)
memory usage: 453.3+ KB
season                  0
date                    0
home_team               0
away_team               0
home_goals              0
away_goals              0
home_xg                 0
away_xg                 0
home_shots              0
away_shots              0
home_shots_on_target    0
away_shots_on_target    0
home_ppda               0
away_ppda               0
home_deep               0
away_deep               0
Result                  0
dtype: int64
```

[34]:

	date	home_goals	away_goals	home_xg
count	3412	3412.000000	3412.000000	3412.000000
mean	2021-03-24 18:22:22.086752768	1.512896	1.210434	1.535249
min	2016-08-12 00:00:00	0.000000	0.000000	0.014500
25%	2018-10-20 00:00:00	1.000000	0.000000	0.864100
50%	2021-03-14 00:00:00	1.000000	1.000000	1.370950
75%	2023-08-12 00:00:00	2.000000	2.000000	2.016350
max	2026-03-22 00:00:00	9.000000	7.000000	5.942400
std	NaN	1.292519	1.154562	0.908507

	away_xg	home_shots	away_shots	home_shots_on_target
count	3412.000000	3412.000000	3412.000000	3412.000000
mean	1.220226	13.349941	11.098769	4.676729
min	0.000000	1.000000	0.000000	0.000000
25%	0.613375	10.000000	8.000000	3.000000
50%	1.069550	13.000000	11.000000	4.000000
75%	1.640725	16.000000	14.000000	6.000000
max	5.067900	37.000000	31.000000	15.000000
std	0.798251	4.926792	4.519015	2.457641

	away_shots_on_target	home_ppda	away_ppda	home_deep
count	3412.000000	3412.000000	3412.000000	3412.000000
mean	3.917644	11.683684	13.059060	6.052462
min	0.000000	2.370400	2.388900	0.000000
25%	2.000000	7.666700	8.377075	3.000000
50%	4.000000	10.384050	11.421100	5.000000
75%	5.000000	14.139800	15.827900	8.000000
max	15.000000	66.833300	85.142900	30.000000
std	2.318135	6.009660	7.005715	4.014179

	away_deep
count	3412.000000
mean	5.086460
min	0.000000
25%	2.000000
50%	4.000000
75%	7.000000
max	30.000000
std	3.707002

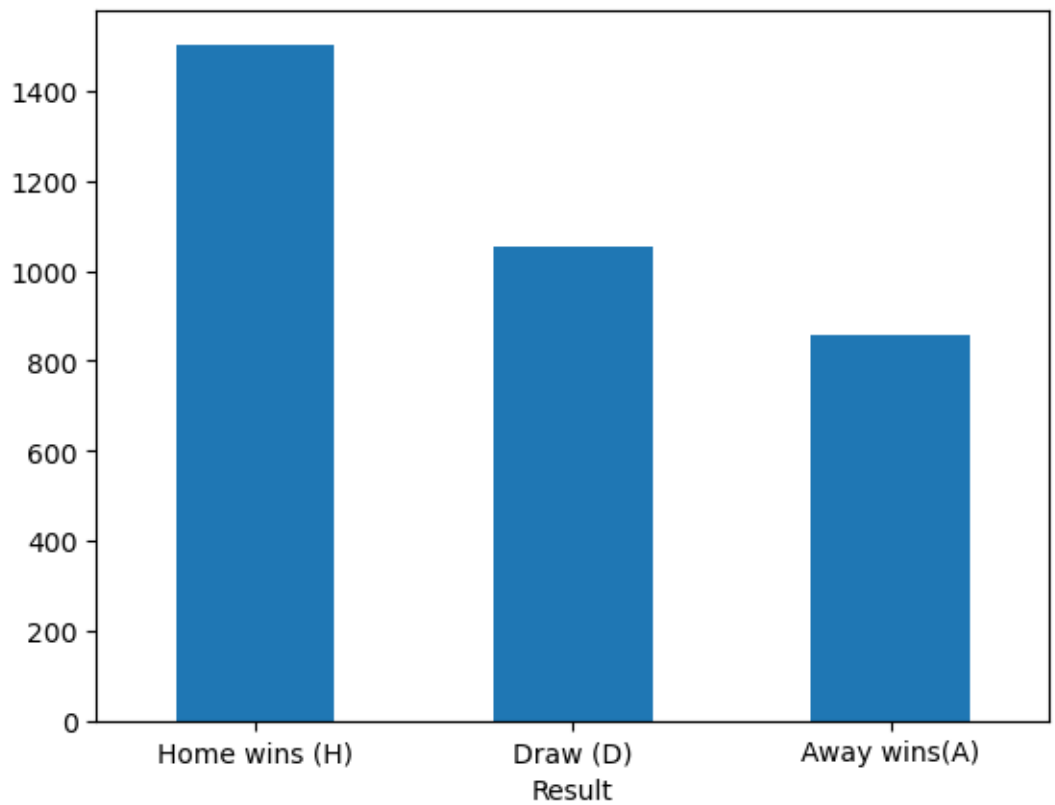
Variables :

- **home_goals** / **away_goals** : number of goals scored by the home / away team
- **home_xg** / **away_xg** : expected goals — sum of the scoring probability of each shot, measures the quality of chances created
- **home_shots** / **away_shots** : total number of shots attempted
- **home_shots_on_target** / **away_shots_on_target** : number of shots on target (that would have required a save from the goalkeeper)

- **home_ppda / away_ppda** : Passes Per Defensive Action - measures pressing intensity ; the lower the value, the higher and more intense the press
- **home_deep / away_deep** : number of passes in the final third — proxy for pressure applied near the opponent's goal

1.1 1. Analysis of home advantage

[35]:
```
counts = df["Result"].value_counts()
counts.plot(kind='bar')
plt.xticks(ticks=[0,1,2], labels=['Home wins (H)', 'Draw (D)', 'Away wins(A)'],␣
 ↪rotation=0)
plt.show()
```

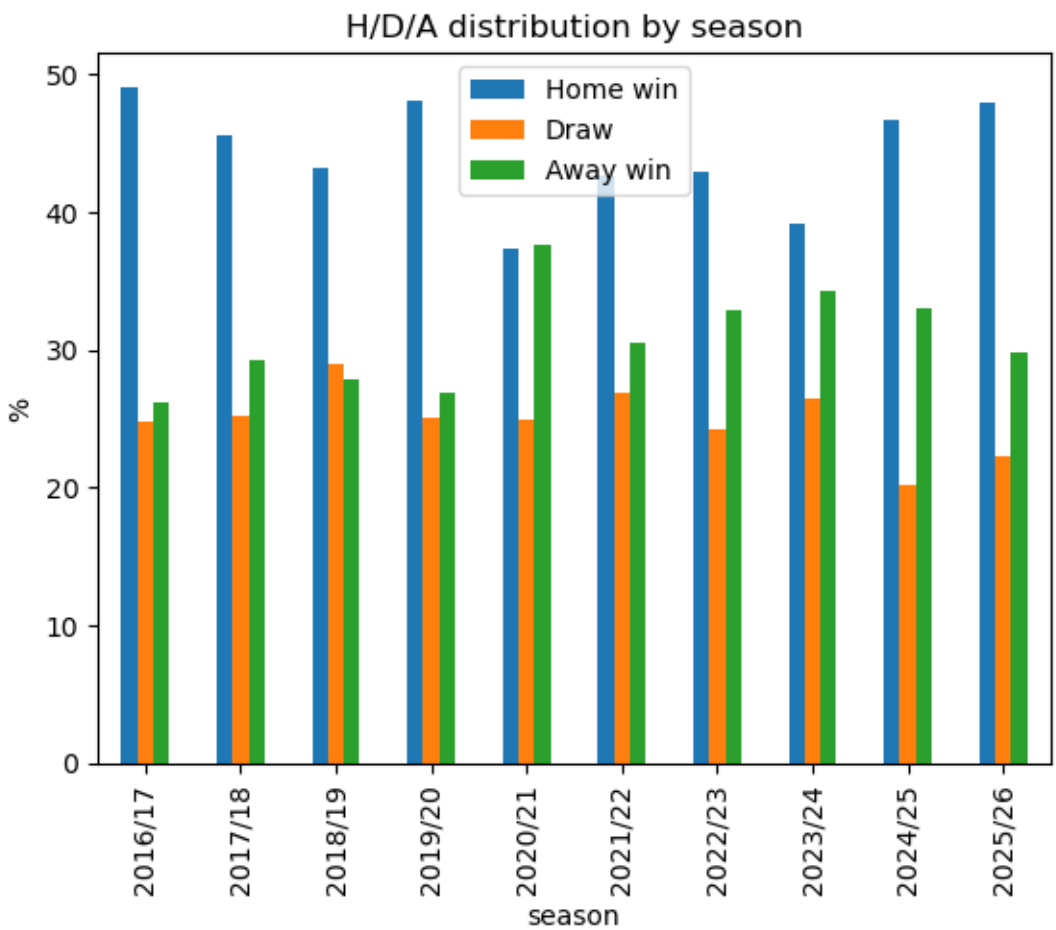


[36]:
```
print("% "+"of Home wins : "+str(round(counts["H"]/len(df["Result"])*100))+"%")
print("% "+"of Away wins : "+str(round(counts["A"]/len(df["Result"])*100))+"%")
print("% "+"of Draws : "+str(round(counts["D"]/len(df["Result"])*100))+"%")
```

```
% of Home wins : 44%
% of Away wins : 31%
% of Draws : 25%
```

[37]:
```
hda_by_season = df.groupby("season")["Result"].value_counts(normalize=True).
↪unstack()*100
hda_by_season[["H", "D", "A"]].plot(kind="bar")
plt.title("H/D/A distribution by season")
plt.ylabel("%")
plt.legend(["Home win", "Draw", "Away win"])
```

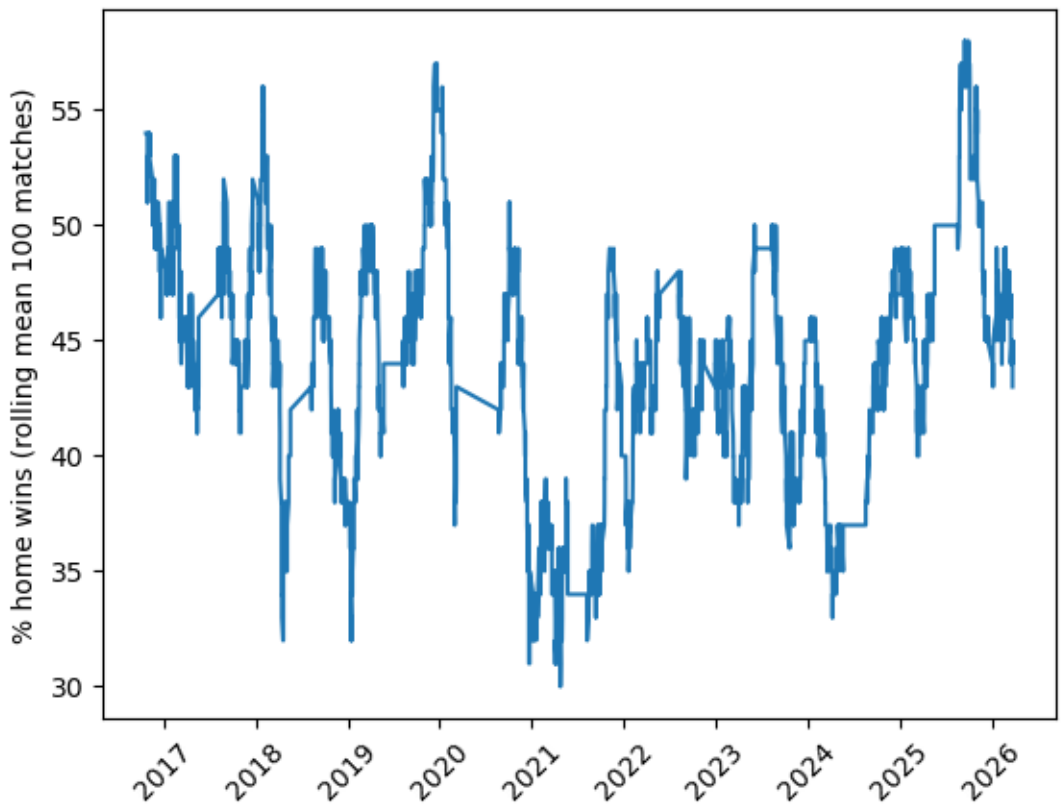
[37]: <matplotlib.legend.Legend at 0x15a461d10>



44% home wins vs. 31% away wins — the home-field advantage is significant in Ligue 1. The goal gap (1.50 vs. 1.31 per match) is more moderate, suggesting that the advantage primarily manifests in tight matches. (seen later)

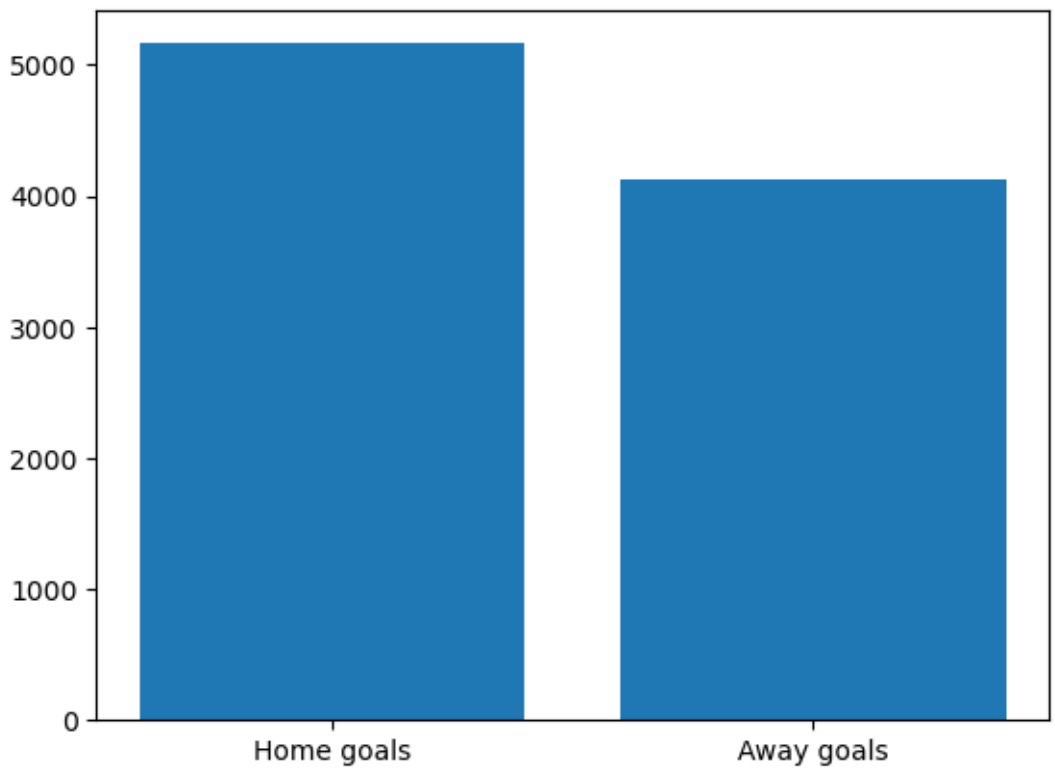
1.1.1 Rolling mean (of 100 matches days) of home wins % during the five seasons :

[38]:
```
df['is_home_win'] = (df['Result'] == 'H').astype(int) #Create a column with 1␣
↪if home win
df['rolling_home_win'] = df['is_home_win'].rolling(window=100).mean() * 100
plt.ylabel("%"+" home wins (rolling mean 100 matches)")
plt.plot(df["date"],df['rolling_home_win'])
plt.xticks(rotation=45)
plt.show()
```



The home win rate fluctuates between 30% and 55%. A notable drop is observed in early 2021 (COVID-19 matches played behind closed doors). The return of fans coincides with a gradual recovery.

[39]:
```
counts =[np.sum(df["home_goals"]),np.sum(df["away_goals"])]
plt.bar([0, 1], counts)
plt.xticks(ticks=[0,1], labels=['Home goals', 'Away goals'], rotation=0)
plt.show()
```



[40]:
```
print("% "+"of Home goals per game : "+str(round(np.sum(df["home_goals"])/
↪len(df["Result"]),2)))
print("% "+"of Away goals per game : "+str(round(np.sum(df["away_goals"])/
↪len(df["Result"]),2)))
```

```
% of Home goals per game : 1.51
% of Away goals per game : 1.21
```

The home advantage is clearer in win rate (+17pp) than in goals per game (+0.19), suggesting that home teams convert close games more effectively rather than dominating scorelines.

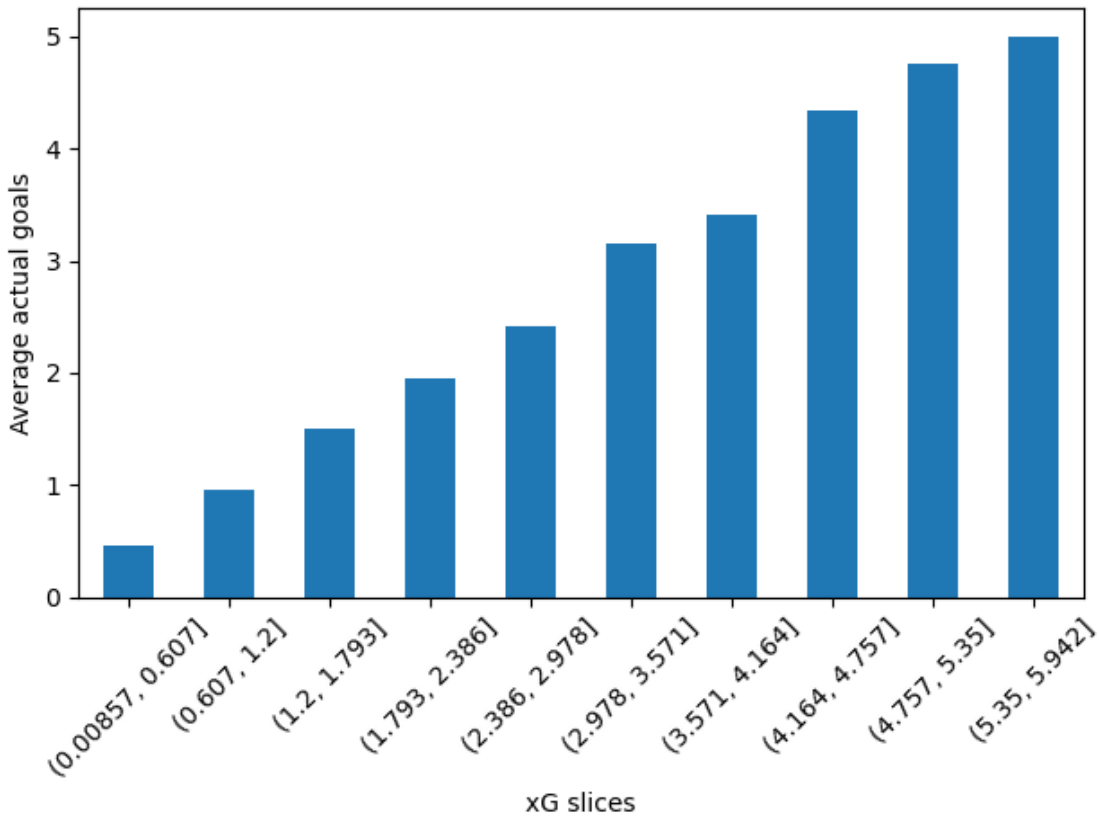
1.2 2. The xG

Let's scatter the actual goals w/ respect to xG

[41]:
```
df['xg_bin'] = pd.cut(df['home_xg'], bins=10) # We cut home_xg column in 10␣
↪slices
avg_goals = df.groupby('xg_bin')['home_goals'].mean()
avg_goals.plot(kind='bar')
plt.xlabel('xG slices')
plt.ylabel('Average actual goals')
plt.xticks(rotation=45)
plt.tight_layout()
```

```
plt.show()
```

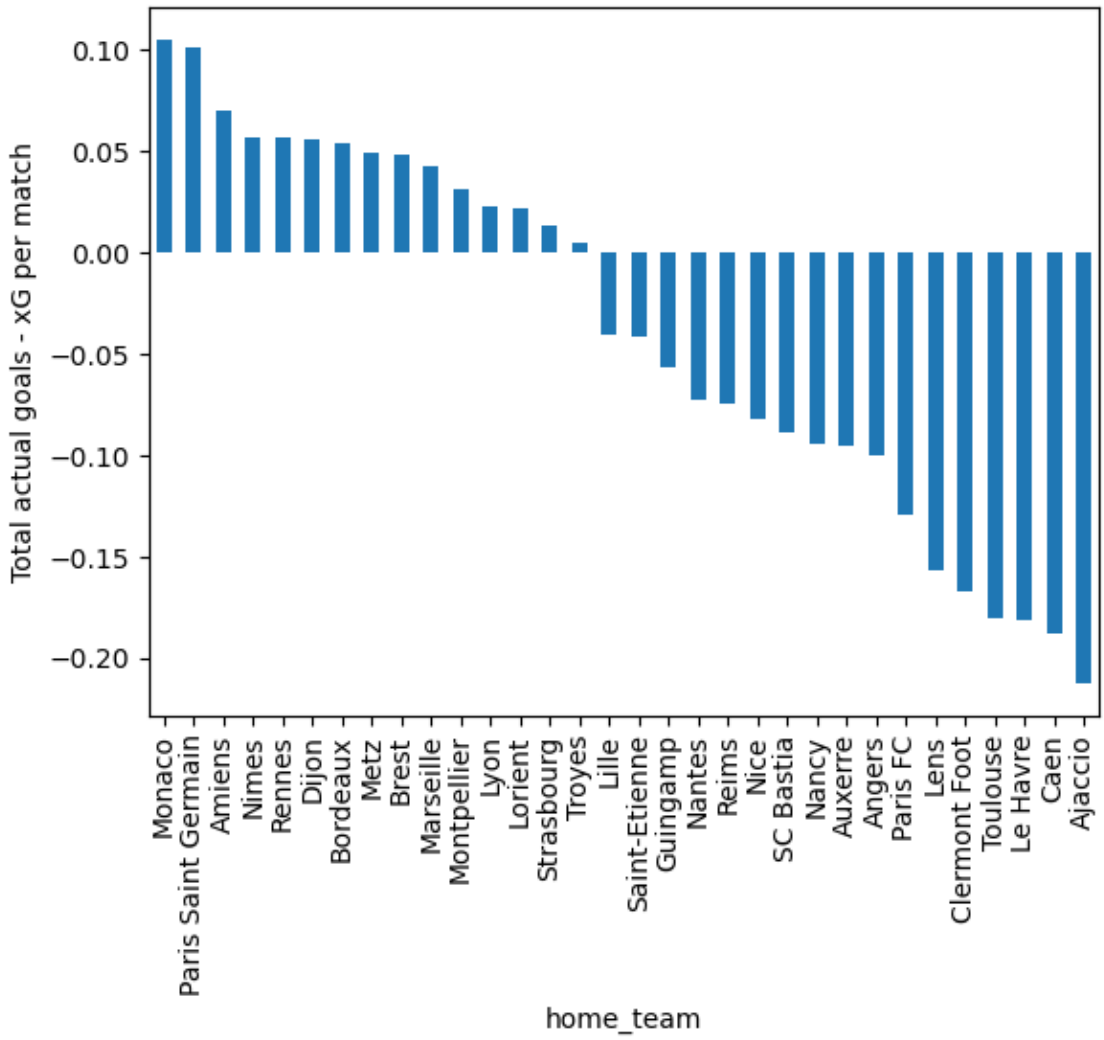
```
/var/folders/wg/p4b819hd79d9_9zf2ts8_wcw0000gn/T/ipykernel_42006/1969410532.py:2
: FutureWarning: The default of observed=False is deprecated and will be changed
to True in a future version of pandas. Pass observed=False to retain current
behavior or observed=True to adopt the future default and silence this warning.
  avg_goals = df.groupby('xg_bin')['home_goals'].mean()
```



The relationship between xG and actual goals is quasi-linear, confirming the relevance of xG as a predictive feature. xG effectively captures the quality of chances created.

[42]:
```
home = df.groupby("home_team")[["home_goals", "home_xg"]].sum()
home["nbr matches"] = df["home_team"].value_counts()
home.columns = ["goals", "xg","nbr_matches"]

away = df.groupby("away_team")[["away_goals", "away_xg"]].sum()
away["nbr_matches"] = df["away_team"].value_counts()
away.columns = ["goals", "xg","nbr_matches"]

team_stats = home.add(away, fill_value=0)
team_stats["diff"] = (team_stats["goals"] - team_stats["xg"])
```

```
team_stats["diff_per_match"] = team_stats["diff"] / team_stats["nbr_matches"]

team_stats.sort_values("diff_per_match", ascending=False)["diff_per_match"].
 ↪plot(kind='bar')
plt.ylabel('Total actual goals - xG per match')
plt.xticks(rotation=90)
plt.show()
```

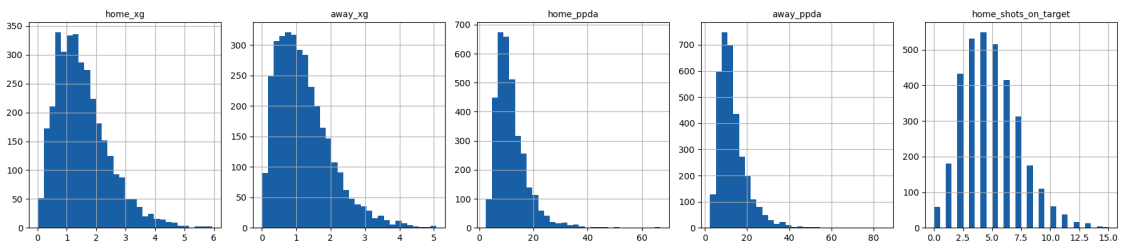


Monaco and PSG are overperforming their xG (quality finishers or luck). Ajaccio and Caen are underperforming (regression to the mean expected). This differential is a valuable feature for identifying teams likely to regress.

1.3 3. Other metrics distribution

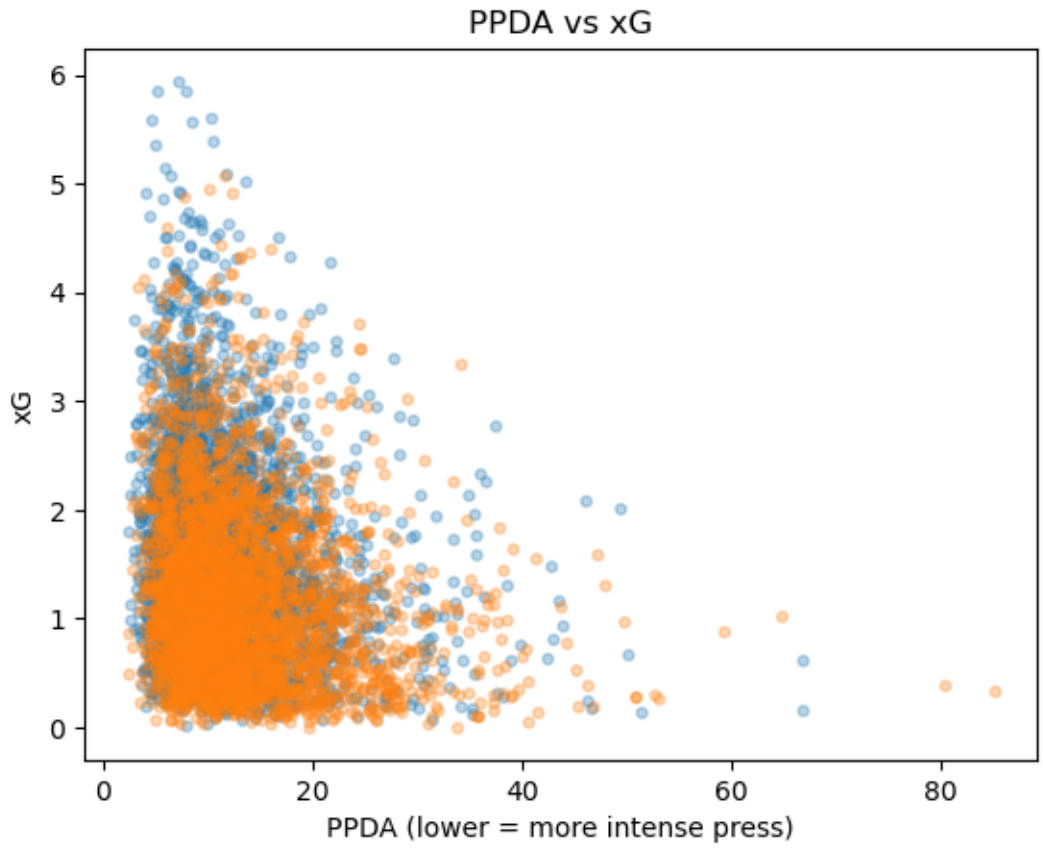
Let's show the distribution of other metrics to have an idea of their value

[43]:
```
features = ['home_xg', 'away_xg', 'home_ppda', 'away_ppda',␣
↪'home_shots_on_target']

fig, axes = plt.subplots(1, 5, figsize=(18, 4))

for ax, col in zip(axes, features):
    df[col].hist(bins=30, ax=ax, color='#185FA5', edgecolor='none')
    ax.set_title(col, fontsize=10)
    ax.set_xlabel('')

plt.tight_layout()
plt.show()
```



[44]:
```
plt.scatter(df['home_ppda'], df['home_xg'], alpha=0.3, s=15, label='Home')
plt.scatter(df['away_ppda'], df['away_xg'], alpha=0.3, s=15, label='Away')
plt.xlabel('PPDA (lower = more intense press)')
plt.ylabel('xG')
plt.title('PPDA vs xG')
```

[44]: Text(0.5, 1.0, 'PPDA vs xG')



1.4 4. Correlation Matrix

Let's compute the correlation for all our metrics

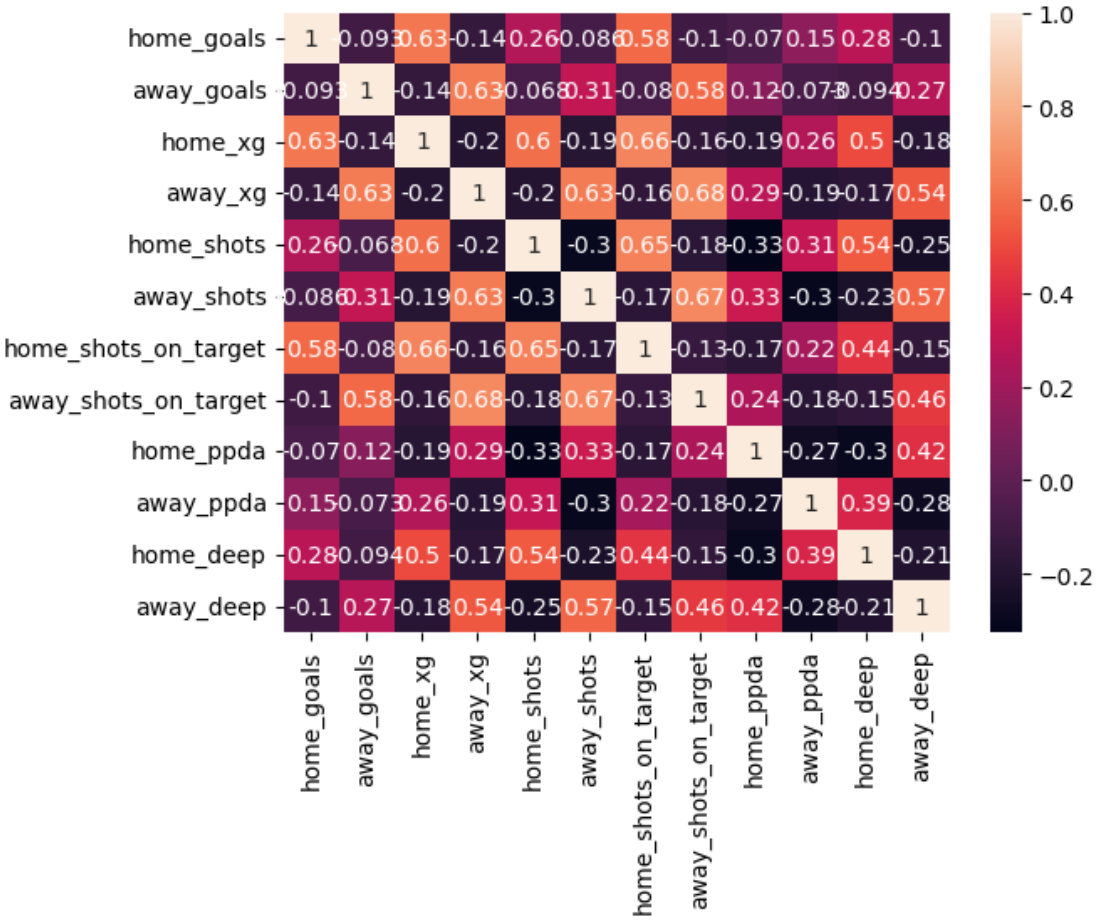
[45]:
```
cols = [
    'home_goals', 'away_goals',
    'home_xg', 'away_xg',
    'home_shots', 'away_shots',
    'home_shots_on_target', 'away_shots_on_target',
    'home_ppda', 'away_ppda',
    'home_deep', 'away_deep'
]

corr_matrix = df[cols].corr()
print(corr_matrix)
```

```
                      home_goals  away_goals   home_xg   away_xg  home_shots  \
home_goals              1.000000   -0.093367  0.626307 -0.143646    0.257703
away_goals             -0.093367    1.000000 -0.141759  0.633345   -0.068200
home_xg                 0.626307   -0.141759  1.000000 -0.199141    0.598864
away_xg                -0.143646    0.633345 -0.199141  1.000000   -0.197975
```

```
home_shots              0.257703   -0.068200  0.598864 -0.197975    1.000000
away_shots             -0.086173    0.307812 -0.188763  0.632804   -0.299538
home_shots_on_target    0.584182   -0.079648  0.662131 -0.155083    0.649688
away_shots_on_target   -0.104977    0.584068 -0.159727  0.676839   -0.179369
home_ppda              -0.069656    0.124804 -0.188728  0.287689   -0.325909
away_ppda               0.146401   -0.073201  0.258005 -0.193178    0.313001
home_deep               0.284117   -0.093535  0.500379 -0.174411    0.541560
away_deep              -0.104098    0.273850 -0.182508  0.538145   -0.248426

                      away_shots  home_shots_on_target  away_shots_on_target  \
home_goals             -0.086173              0.584182             -0.104977
away_goals              0.307812             -0.079648              0.584068
home_xg                -0.188763              0.662131             -0.159727
away_xg                 0.632804             -0.155083              0.676839
home_shots             -0.299538              0.649688             -0.179369
away_shots              1.000000             -0.173008              0.671314
home_shots_on_target   -0.173008              1.000000             -0.131006
away_shots_on_target    0.671314             -0.131006              1.000000
home_ppda               0.331899             -0.172590              0.239387
away_ppda              -0.301469              0.221808             -0.184584
home_deep              -0.229017              0.436091             -0.153785
away_deep               0.573911             -0.152486              0.455867

                      home_ppda  away_ppda  home_deep  away_deep
home_goals            -0.069656   0.146401   0.284117  -0.104098
away_goals             0.124804  -0.073201  -0.093535   0.273850
home_xg               -0.188728   0.258005   0.500379  -0.182508
away_xg                0.287689  -0.193178  -0.174411   0.538145
home_shots            -0.325909   0.313001   0.541560  -0.248426
away_shots             0.331899  -0.301469  -0.229017   0.573911
home_shots_on_target  -0.172590   0.221808   0.436091  -0.152486
away_shots_on_target   0.239387  -0.184584  -0.153785   0.455867
home_ppda              1.000000  -0.267309  -0.296150   0.420094
away_ppda             -0.267309   1.000000   0.386571  -0.284124
home_deep             -0.296150   0.386571   1.000000  -0.207190
away_deep              0.420094  -0.284124  -0.207190   1.000000
```

[46]:
```
sns.heatmap(corr_matrix,annot=True)
```

[46]:
```
<Axes: >
```



The variables most highly correlated with goals are xG (0.63) and shots on target (0.58). PPDA and deep completions show weaker but non-negligible correlations (~0.1-0.27). Note: shots and xG are highly correlated with each other (0.62); this multicollinearity will need to be addressed during modeling.